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Economic Report and Production Analysis for Japan

Heyang Huang, Soulaymane Kachani, Jiayu Zhao
COLUMBIA UNIVERSITY IN THE CITY OF NEW YORK

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0. Executive Summary

In this report, we aim to uncover the economic path Japan went through during the period from 2009 to 2014. Specifically, we dive into the topic through a macroeconomic overview of Japan from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We find that during the period of our interest, Japan demonstrated a particularly low level of economic growth with an annual GDP growth rate of -1.5%, concurrent with a stagnation in economic complexity as compared with other major economies in Asia.

The lack of economic progress during the period, we believe, can be partially explained through analysis of the economic and product complexity structure of Japan on the industry group level. The product classification system we adopt allow us to measure the actual development for various industry groups and to observe that Japan aggressively invested in high-tech, machinery and transportation industry groups during the period, while its developed peers and emerging economies around the globe have adopted a slightly more conservative growth path.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, Japan could further its gains by staying optimal with its current industry structure and there is no immediate need of reform because no industry has a positive opportunity gain at its current stage. We further note that Japan has a slightly mismatched development strategy that indicates a moderate level of structural optimality on the complexity level, having an adjusted R-square of around 58.3%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Japan indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure balancing as suggested in the paper.

1. Report Overview

In this report, we aim to uncover the economic path Japan went through during the period from 2009 to 2014. Specifically, we dive into the topic through a macroeconomic overview of Japan from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We then analyze the industrial framework and changes of Japan during the period based on quantitative measures such as Product Complexity Index (PCI) and Product Family Complexity Index (PfCI), followed by a detailed analysis of how such changes have matched with development strategies as quantitatively measured by opportunity gains on an industry-level. We conclude our report with macro-level results and predictions of Japan, as deduced from regression models we have established on a global scale, which takes into account countries' stances in their product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

As the 3rd largest economy in the world by GDP, Japan has a GDP of 5.7 trillion USD in 2009 and 4.9 trillion USD in 2014, exhibiting an extraordinary level of economic output. Figure 1 illustrates the time-series GDP results for world's leading economies from 1996-2016, and Japan ranks among its counterparts as a stable player. Japan exhibitsa consistent level of economic growth during the twenty years from 1996 to 2016, but from 2009 to 2014, Japan has an annual GDP growth rate of -1.5%. This rate is clearly below the annual growth rate of the group of leading economies in the world, which have a geometric average growth rate of 5.1% and an arithmetic average growth rate of 5.2%. The annual growth rate of -1.5% is particularly in sharp contrast to an annual growth rate of 15.5% for China, one of Japan's major competitors in the market. The stagnated economic growth in Japan, we believe, can be reflected through consistent ECI and PCI, and can be further explained by a mismatched set of production strategies Japan adopted during the period, which we will expand at the later stage of the report.

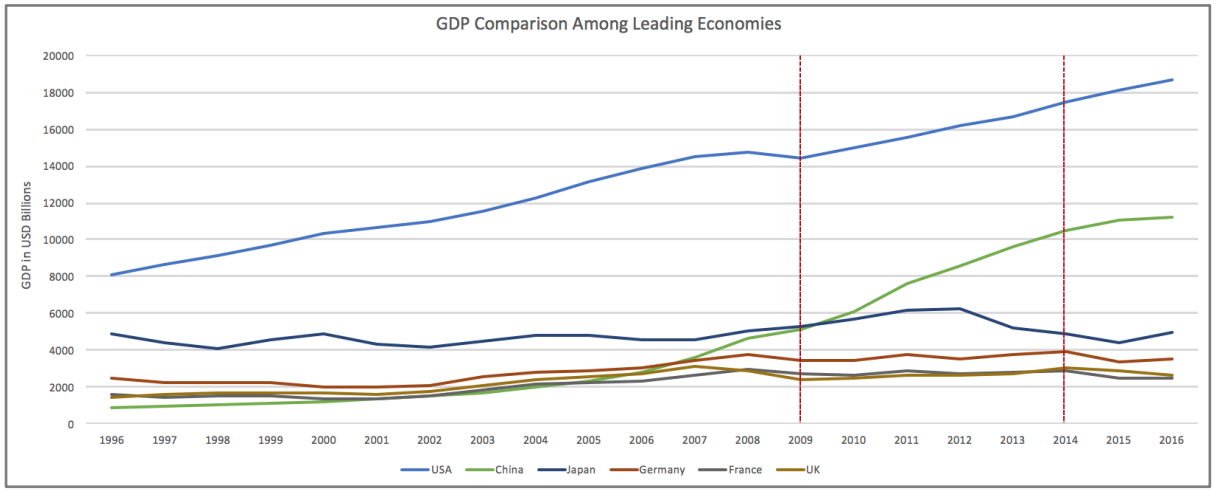


Figure 1: GDP Comparison Among Leading Economies

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), as defined by Ricardo Hausmann and Cesar Hidalgo in 2009 to explain the knowledge accumulated in a country's population, exhibits similar pattern for Japan. As illustrated in Figure 2 below, Japan exhibits a stable but flat curve in terms of its production capability from 2009 to 2014, which is true for most of the developed economies. This is in direct contrast to countries such as China, which demonstrates overwhelming advantage over Japan in terms of both ECI and GDP growth during the period.

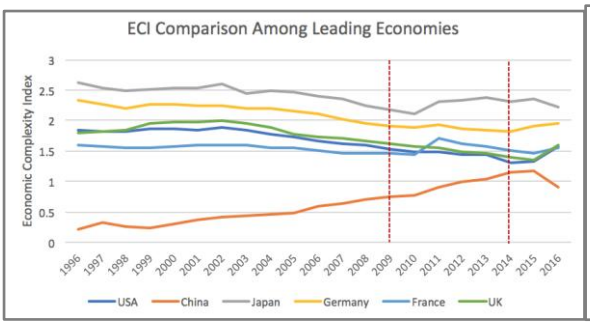


Figure 2: ECI Comparison

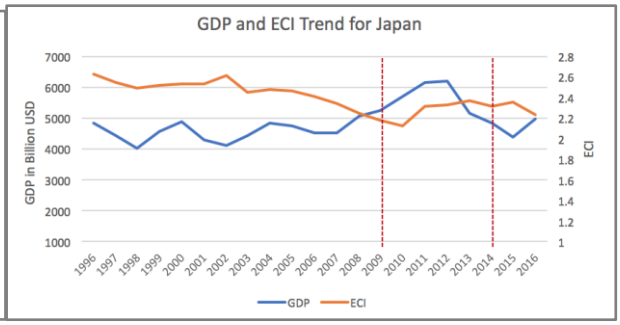


Figure 3: GDP and ECI (Japan)

Figure 3 offers a further comparison of GDP and ECI over the entire time span from 1996 to 2016 for Japan. The stagnancy in GDP during the period from 2009 to 2014 is accompanied by a stagnancy in ECI, indicting the potential that the mild economic growth experienced by Japan during the period from 2009 to 2014 might be attributed to its slow-growing ECI, or the ineffectiveness of the country in developing a competitive product space that helps to accumulate complicated knowledge and technology. In order to provide a more concrete answer to why this has happened, we will dive into an analysis of Japan's production structure and its product space designs in the next few chapters.

3. ProductionAnalysis

3.1 Product Categorization

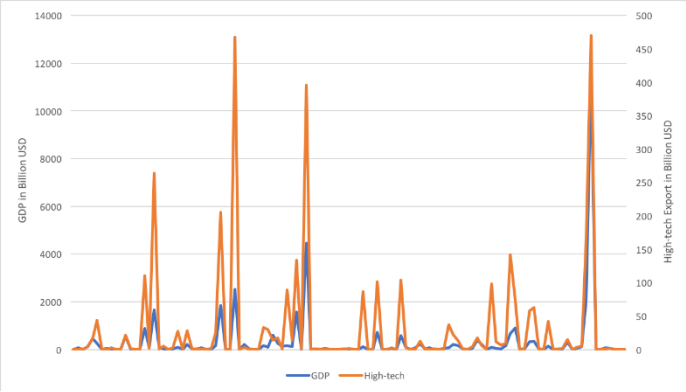
Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Product in our data source are labelled from 010000-999999, with the first two-digits representing a general category to which the product belongs. Based on the 99 categories as extracted from the label, we further group the products into 18 most widely-accepted industries, as summarized in Table 1.

Note that the high-tech industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technical gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), high-tech industry has the highest PfCI among all industry groups.

Category	Description
Animal and By-product	All kinds of meat, livestock and related by-products
Vegetable, Fruit and Plant	All kinds of plants, edible vegetables and fruits
Food and Preparation	Edible preparations, raw cooking materials and appliances
Mineral and Stone	Mining industry for mineral and stone materials
Chemical	Raw chemical materials and by-products
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Wood Product	Book, paper and other elementary products related to wood
Clothing and Accessory	All kinds of fabrics, filament, clothing and accessories
Textile	All kinds of textile articles and related products
Metal	Mining of all sorts of raw metal
Machinery	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapon	Explosives, arms and ammunition
Daily Instrument	Basic daily instruments such as sport requisites, timing and measuring tools
Precious Stone	Refined or processed stone products such as pearl and ceramics
Service	Financial, business, legal, communication, travel services, etc.
High-tech	Agglomeration of advanced products with highest PCIs

Table 1: Categorization of Products

We further observe a strong positivecorrelation between the GDP and the export of high-tech products. As shown by Figure 4 below, a high adjusted R-square of 67.6% exists between GDP and high-tech exports, indicating that more successful economics, as measure by GDP, tend to be those that are capable of producing and exporting high-tech products as we define them.



Statistics	Value
Multiple R	0.824
R-Square	0.679
Adjusted R^2	0.676
Variable	**Value**
Coefficient	12.20
t-stat	15.61
P-value	0(***)

Figure 4: Regression of GDP and High-tech Exports

3.2 Actual Development

Based on the product categorization aforementioned, we aim to capture Japan's actual production strategy between 2009 and 2014 on a complexity level.In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is in developing products within that group. Defining this net distance change as the Actual Developmental Index (ADI), our export data allows us to calculate and compare ADIofJapan and the 18 product groups in 2009 and 2014 through subtraction, where a positive ADI

indicates a shortening of the distance (i.e., advancement of the industry), and a negative ADI indicates the reverse. As ranked in Table 2, during the period from 2009to 2014, Japan has focused on products with a high level of complexity and those requiring advancedskills, including high-tech, machinery and transportation, as is unusualamongAsian economies.

Category	ADI	Category	ADI
1. High-tech	0.115	2. Machinery	0.103
3. Transportation	0.090	4. Mineral and Stone	0.087
5. Weapon	0.087	6. Food and Preparation	0.074
7. Wood Product	0.067	8. Service	0.067
9. Vegetable, Fruit and Plant	0.065	10. Metal	0.062
11. Chemicals	0.060	12. Clothing and Accessory	0.059
13. Textile	0.044	14. Precious Stone	0.043
15. Animal and By-product	0.040	16. Chemical Manufacturing	0.016
17. Daily Instrument	0.015	18. Raw Manufacturing	-0.001

Table 2: Actual Developmental Index (ADI) of Japan

3.3 Regional and Global Comparison

Table 3and 4 below summarizes how the production strategy adopted by Japan compares with its Asian counterparts and a selected group of leading economies in the world, as measured in ADI. It can be directly observed that Japan focused on products with a higher complexity level, when compared against its Asian counterparts such as China and India, which give a considerable level of attention to more basic product groups such as precious stone and daily instrument.

Ranking	Japan	China	Korea	India
1	High-tech	Precious Stone	Chemicals	Daily Instrument
2	Machinery	Service	Machinery	High-tech
3	Transportation	High-tech	High-tech	Wood Product
4	Mineral and Stone	Machinery	Weapon	Machinery
5	Weapon	Chemicals	Precious Stone	Chemicals

Table 3: ADIComparison on a Regional Scale

Ranking	USA	Germany	Singapore	France
1	Service	Chemicals	Machinery	Raw Manufacturing
2	Chemical Manufacturing	Service	Transportation	Mineral and Stone
3	Food and Preparation	High-tech	Daily Instrument	Precious Stone
4	Textile	Transportation	Chemicals	Chemical Manufacturing
5	Clothing and Accessory	Food and Prep	High-tech	Chemicals

Table 4: ADI Comparison on a Global Scale

Table 4 further examines Japan’s development strategy against those of more developed economies on a global scale. Reasonably, developed economics such as USA, Germany, Singapore and Japan have devoted the majority of their efforts into highly complicated industry groups while emerging economies such as Japan still partially relies on basic industry groups like mineral and stone. This trend holds true in general, regardless of the continent or the time period, and serves as an indicator for which phase of development a country is currently in as evaluated from their production strategy. Based on this standard,

Japan is indeed adopting an advanced and ambitious strategy that seeks development in industry groups with ahigh level of complexity.

4. Product Space Design

4.1 Suggested Development Vector

As aforementioned, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined inHausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP=0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter in the product as compared to global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in term of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP=1) has an opportunity gain of zero. And unreasonable investment on certain product p would lead to a relative decrement in export of more profitable products, and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economics strength, those having higher ECI, or in other words, more reasonable economics structure is often in better shape.

Ranking	Mexico	Japan	South Korea	Germany	China	Malaysia	France
1	Vegetable, Fruit and Plant	Daily Instrument	Chemicals	Mineral and Stone	Mineral and Stone	Raw Manufacturing	Service
2	Daily Instrument	Transportation	Daily Instrument	Chemicals	Clothing and Accessory	Daily Instrument	High-tech
3	Transportation	Machinery	Transportation	Daily Instrument	Chemicals	Vegetable, Fruit and Plant	Food and Preparation
4	Machinery	Metal	Machinery	Transportation	Daily Instrument	Machinery	Chemical Manufacturing
5	High-tech	High-tech	Metal	Machinery	Machinery	Mineral and Stone	Raw Manufacturing
6	Weapon	Chemicals	Textile	Metal	Metal	High-tech	Animal and By-product
7	Clothing and Accessory	Weapon	High-tech	High-tech	Textile	Chemicals	Transportation
8	Chemical Manufacturing	Mineral and Stone	Raw Manufacturing	Wood Product	High-tech	Precious Stone	Mineral and Stone
9	Chemicals	Chemical Manufacturing	Mineral and Stone	Raw Manufacturing	Wood Product	Weapon	Clothing and Accessory
10	Textile	Clothing and Accessory	Weapon	Chemical Manufacturing	Raw Manufacturing	Animal and By-product	Chemicals
11	Service	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Service	Weapon
12	Precious Stone	Textile	Clothing and Accessory	Weapon	Weapon	Clothing and Accessory	Textile
13	Animal and By-product	Service	Service	Clothing and Accessory	Service	Food and Preparation	Vegetable, Fruit and Plant
14	Mineral and Stone	Animal and By-product	Chemical Manufacturing	Service	Animal and By-product	Metal	Precious Stone
15	Food and Preparation	Wood Product	Animal and By-product	Animal and By-product	Vegetable, Fruit and Plant	Chemical Manufacturing	Wood Product
16	Wood Product	Raw Manufacturing	Wood Product	Vegetable, Fruit and Plant	Chemical Manufacturing	Textile	Metal
17	Raw Manufacturing	Food and Preparation	Vegetable, Fruit and Plant	Textile	Food and Preparation	Wood Product	Daily Instrument
18	Metal	Vegetable, Fruit and Plant	Food and Preparation	Food and Preparation	Transportation	Transportation	Machinery
EfCI	2.901	2.491	2.117	1.910	1.591	1.772	1.224

Table 5: SDV of High E_fCI Countries 2014

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 shows the SDV ofa selected group of leading economies with outstanding economic complexity, with the industry groups ranked in decreasing order of significance. Dark grey indicates positive opportunity gain, gray indicates zero opportunity gains, and light grey indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development.In contrast, countries listed in Table 6 have nearly half of industries with positive opportunity gain. While the economies with absolute negative E_fCI, such as Nigeria and Algeria, all need to develop High-tech, machinery and daily instruments, those with median E_fCI are facing a more complicated and diverse situation, such as Italy, Spain and USA.

It can be concluded that Japan is categorized to be a highly complexed country as compared with the world's leading economies and other major economies in Europe.

Ranking	Italy	Spain	USA	Nigeria	Algeria
1	Mineral and Stone	Mineral and Stone	Machinery	High-tech	High-tech
2	Chemicals	Precious Stone	High-tech	Machinery	Machinery
3	Vegetable, Fruit and Plant	Clothing and Accessory	Daily Instrument	Daily Instrument	Daily Instrument
4	Service	Chemicals	Transportation	Transportation	Transportation
5	High-tech	Textile	Textile	Textile	Textile
6	Food and Preparation	Daily Instrument	Raw Manufacturing	Chemical Manufacturing	Chemical Manufacturing
7	Chemical Manufacturing	High-tech	Service	Raw Manufacturing	Raw Manufacturing
8	Raw Manufacturing	Machinery	Weapon	Metal	Metal
9	Wood Product	Chemical Manufacturing	Chemical Manufacturing	Chemicals	Mineral and Stone
10	Clothing and Accessory	Vegetable, Fruit and Plant	Chemicals	Mineral and Stone	Chemicals
11	Textile	Food and Preparation	Metal	Weapon	Weapon
12	Metal	Service	Clothing and Accessory	Wood Product	Wood Product
13	Machinery	Raw Manufacturing	Wood Product	Clothing and Accessory	Animal and By-product
14	Transportation	Wood Product	Animal and By-product	Animal and By-product	Clothing and Accessory
15	Weapon	Metal	Food and Preparation	Precious Stone	Service
16	Daily Instrument	Transportation	Precious Stone	Service	Food and Preparation
17	Precious Stone	Weapon	Vegetable, Fruit and Plant	Food and Preparation	Precious Stone
18	Animal and By-product	Animal and By-product	Mineral and Stone	Vegetable, Fruit and Plan	Vegetable, Fruit and Plant
EfCI	1.271	0.371	0.285	-1.239	-2.434

Table 6: SDV of Other Typical Countries 2014

4.2 Sensitivity Analysis and Contingency Strategy

Japan has a characteristic SDV of high-complexity economy and is embedded with massive production knowledge. Japan should take precautions against unexpected strikes industry by industry and forecast the adverse effects. The regression of an industry with low PCI and low Opportunity Gain is less dangerous than that of cornerstone industries. High-tech is such a representative of the cornerstone industries, because High-tech connects to most of the profitable products that developed countries are currently specializing in. For the same token, Food and Preparation is one of the less fundamental industries for developed countries. Such intuition can be rigorously proved by the sensitivity analysis we are about to implement.

The sensitivity analysis aims to approximate the volatility of SDV subject to aninstant increment or decrement of export and is mainly implemented on developed countries thathave few industries with positive opportunity gains. To be more specific, sensitivity score is theleast-squared slope for the linear regression between percentage change of production for
Country C and Product P and the E_f CI of C.

When the opportunity gain of the industry is negative, the sensitivity analysis approximates how an over-development of the industry jeopardizes the economic structure of a country. When the opportunity gain of the industry is zero, the sensitivity analysis determines whether the further development of the industry shapes the country a better economic structure, and predicts how the SDV would change accordingly. Specifically, as we reduce the export of such an industry to 25%-50% of its original value, a few other industries originally with negative opportunity gain would probably become profitable. We separate the increment and decrement of production in the sensitivity analysis, because they tend to show vast difference. The considerable profitability of a production spike doesn't imply a shrink is always followed by a painful crisis.

We perform the sensitivity analysis of the aggregate of top ECI countries, including US, UK, Korea, Japan, South Korea, Singapore, Germany, and France, and use the result as the benchmark. The sensitivity score in 2014 and 2009 are shown in Table 7 and Table 8 respectively:

	High-tech	Food and Preparation
Decremental	1.676	0.007
Incremental	1.195	-0.006

Table 7: Sensitivity Score of the Aggregate of Developed Countries 2014

	High-tech	Food and Preparation
Decremental	1.514	-0.001
Incremental	1.294	0.007

Table 8: Sensitivity Score of the Aggregate of Developed Countries 2009

In summary, High-tech exerts great impacts on economics structure of top countries. Such countries are extremely vulnerable to High-tech industry declines, and benefit from High-tech prosperity. In contrast, ECI is not highly sensitive to shift of Food and Preparation industry.

We observe a similar pattern that the Sensitivity Score of Food and Preparation is approximately zero, while the Decremental Sensitivity Score is larger than the Increment. We can therefore make three conclusions here.

1. The disturbance in Food and Preparation industry exert a trivial impact on the E_fCI of a country, which echoes with our intuition that Food and Preparation is of minimal significance to the structure of product space of developed countries or fast developing countries.
2. The High-tech industry is the cornerstone industry of most developed or fast developing countries.
3. The reward of an increase of High-tech production is less than the jeopardy of a decrease in the same extent.

Besides the sensitivity score, we are interested in the shift of SDV caused by upheavals in different industries. The shift of SDV gives us a good vision of how the production pattern will be affected. The results of Benchmark in 2014 are shown below in Table 9-Table 16

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
2	Raw Manufacturing	Chemicals	Chemicals	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
3	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
4	Transportation	Transportation	Transportation	Transportation	Transportation	Transportation
5	Machinery	Machinery	Machinery	Machinery	Machinery	Machinery
6	Textile	Metal	Metal	Metal	Metal	Metal
7	High-tech	Textile	Textile	Textile	Textile	Textile
8	Weapon	High-tech	High-tech	Chemicals	Chemicals	Chemicals
9	Chemicals	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
10	Precious Stone	Mineral and Stone	Chemical Manufacturing	Mineral and Stone	Mineral and Stone	Mineral and Stone
11	Service	Weapon	Mineral and Stone	Weapon	Weapon	Weapon
12	Chemical Manufacturing	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
13	Animal and By-product	Service	Weapon	Service	Service	Service
14	Mineral and Stone	Animal and By-product	Service	Animal and By-product	Animal and By-product	Animal and By-product
15	Wood Product	Chemical Manufacturing	Animal and By-product	Wood Product	Wood Product	Wood Product
16	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Metal	Wood Product	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation
18	Food and Preparation	Food and Preparation	Wood Product	High-tech	High-tech	High-tech
EfCI	2.469	1.978	1.886	1.593	1.594	1.602

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Table 9: Aggregated High-tech 2014(Decremental Sensitivity)

Ranking	0	"+50%"	"+100%"	"+150%"	"+200%"	"+250%"
1	Clothing and Accessory	Daily Instrument	High-tech	Daily Instrument	Daily Instrument	Daily Instrument
2	Raw Manufacturing	Transportation	Daily Instrument	Machinery	Machinery	Machinery
3	Daily Instrument	Machinery	Transportation	Transportation	Transportation	Transportation
4	Transportation	Textile	Machinery	High-tech	High-tech	High-tech
5	Machinery	High-tech	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
6	Textile	Weapon	Weapon	Textile	Textile	Textile
7	High-tech	Chemicals	Chemicals	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
8	Weapon	Chemical Manufacturing	Textile	Chemicals	Weapon	Weapon
9	Chemicals	Clothing and Accessory	Clothing and Accessory	Weapon	Chemicals	Chemicals
10	Precious Stone	Precious Stone	Service	Metal	Metal	Clothing and Accessory
11	Service	Service	Raw Manufacturing	Clothing and Accessory	Clothing and Accessory	Metal
12	Chemical Manufacturing	Metal	Precious Stone	Wood Product	Wood Product	Wood Product
13	Animal and By-product	Animal and By-product	Metal	Service	Service	Service
14	Mineral and Stone	Raw Manufacturing	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
15	Wood Product	Wood Product	Wood Product	Food and Preparation	Food and Preparation	Food and Preparation
16	Vegetable, Fruit and Plant	Mineral and Stone	Food and Preparation	Precious Stone	Precious Stone	Precious Stone
17	Metal	Food and Preparation	Mineral and Stone	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
18	Food and Preparation	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Mineral and Stone	Mineral and Stone	Mineral and Stone
EfCI	2.469	3.317	3.965	5.592	5.766	5.717

Table 10: Aggregated High-tech 2014(Incremental Sensitivity)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
2	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
3	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
4	Transportation	Transportation	Transportation	Transportation	Transportation	Transportation
5	Machinery	Machinery	Machinery	Machinery	Machinery	Machinery
6	Textile	Textile	Textile	Textile	Textile	Textile
7	High-tech	High-tech	High-tech	High-tech	High-tech	High-tech
8	Weapon	Weapon	Weapon	Weapon	Weapon	Weapon
9	Chemicals	Chemicals	Chemicals	Chemicals	Chemicals	Chemicals
10	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
11	Service	Service	Service	Service	Service	Service
12	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
13	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
14	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
15	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product
16	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Metal	Metal	Metal	Metal	Metal	Metal
18	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation
EfCI	2.469	2.469	2.469	2.465	2.465	2.465

Table 11: Aggregated Food and Preparation 2014(Decremental Sensitivity)

Ranking	0	"+50%"	"+100%"	"+150%"	"+200%"	"+250%"
1	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
2	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
3	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
4	Transportation	Transportation	Transportation	Transportation	Transportation	Transportation
5	Machinery	Machinery	Machinery	Machinery	Machinery	Machinery
6	Textile	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation
7	High-tech	Textile	Textile	Textile	Textile	Textile
8	Weapon	High-tech	High-tech	High-tech	High-tech	High-tech
9	Chemicals	Weapon	Weapon	Weapon	Weapon	Weapon
10	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
11	Service	Service	Mineral and Stone	Service	Service	Service
12	Chemical Manufacturing	Mineral and Stone	Service	Mineral and Stone	Mineral and Stone	Mineral and Stone
13	Animal and By-product	Chemicals	Chemicals	Chemicals	Chemicals	Chemicals
14	Mineral and Stone	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
15	Wood Product	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
16	Vegetable, Fruit and Plant	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
17	Metal	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product
18	Food and Preparation	Metal	Metal	Metal	Metal	Metal
EfCI	2.469	2.100	2.099	2.098	2.098	2.098

Table 12: Aggregated Food and Preparation 2014(Incremental Sensitivity)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Service	Service	Service
2	High-tech	Raw Manufacturing	Raw Manufacturing	Metal	Metal	Metal
3	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemicals	Chemicals	Chemicals
4	Daily Instrument	Daily Instrument	Daily Instrument	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
5	Weapon	Weapon	Weapon	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
6	Transportation	Transportation	Transportation	Textile	Textile	Textile
7	Machinery	Machinery	Machinery	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
8	Raw Manufacturing	Metal	Metal	Machinery	Machinery	Machinery
9	Textile	Textile	Textile	Transportation	Transportation	Transportation
10	Service	High-tech	High-tech	Weapon	Weapon	Weapon
11	Chemicals	Service	Service	Daily Instrument	Daily Instrument	Daily Instrument
12	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
13	Animal and By-product	Chemicals	Chemicals	Animal and By-product	Animal and By-product	Animal and By-product
14	Mineral and Stone	Animal and By-product	Animal and By-product	Mineral and Stone	Mineral and Stone	Mineral and Stone
15	Wood Product	Mineral and Stone	Mineral and Stone	Wood Product	Wood Product	Wood Product
16	Vegetable, Fruit and Plant	Wood Product	Wood Product	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Metal	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Food and Preparation	Food and Preparation	Food and Preparation
18	Food and Preparation	Food and Preparation	Food and Preparation	High-tech	High-tech	High-tech
EfCI	1.911	1.767	1.767	1.204	1.204	1.245

Table 13: Aggregated High-tech 2009 (Decremental Sensitivity)

Ranking	0	"+50%"	"+100%"	"+150%"	"+200%"	"+250%"
1	Clothing and Accessory	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
2	High-tech	Transportation	Transportation	Transportation	Transportation	Transportation
3	Chemical Manufacturing	Machinery	Machinery	Machinery	Machinery	Machinery
4	Daily Instrument	Textile	High-tech	High-tech	High-tech	High-tech
5	Weapon	High-tech	Chemicals	Chemicals	Chemicals	Chemicals
6	Transportation	Chemicals	Weapon	Weapon	Weapon	Weapon
7	Machinery	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Textile	Textile
8	Raw Manufacturing	Weapon	Textile	Textile	Chemical Manufacturing	Chemical Manufacturing
9	Textile	Clothing and Accessory	Mineral and Stone	Mineral and Stone	Raw Manufacturing	Raw Manufacturing
10	Service	Mineral and Stone	Metal	Metal	Metal	Metal
11	Chemicals	Precious Stone	Raw Manufacturing	Raw Manufacturing	Mineral and Stone	Mineral and Stone
12	Precious Stone	Metal	Clothing and Accessory	Service	Wood Product	Wood Product
13	Animal and By-product	Service	Service	Clothing and Accessory	Service	Precious Stone
14	Mineral and Stone	Raw Manufacturing	Precious Stone	Precious Stone	Clothing and Accessory	Clothing and Accessory
15	Wood Product	Wood Product	Wood Product	Wood Product	Precious Stone	Service
16	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Metal	Animal and By-product	Animal and By-product	Animal and By-product	Food and Preparation	Food and Preparation
18	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Animal and By-product	Animal and By-product
EfCI	1.911	3.474	4.237	4.297	6.197	6.368

Table 14: Aggregated High-tech 2009 (Incremental Sensitivity)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
2	High-tech	High-tech	High-tech	High-tech	High-tech	High-tech
3	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
4	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
5	Weapon	Weapon	Weapon	Weapon	Weapon	Weapon
6	Transportation	Transportation	Transportation	Transportation	Transportation	Transportation
7	Machinery	Machinery	Machinery	Machinery	Machinery	Machinery
8	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
9	Textile	Textile	Textile	Textile	Textile	Textile
10	Service	Service	Service	Service	Service	Service
11	Chemicals	Chemicals	Chemicals	Chemicals	Chemicals	Chemicals
12	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
13	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
14	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
15	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product
16	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Metal	Metal	Metal	Metal	Metal	Metal
18	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation
EfCI	1.911	1.911	1.912	1.912	1.912	1.912

Table 15: Aggregated Food and Preparation 2009 (Decremental Sensitivity)

Ranking	0	"+50%"	"+100%"	"+150%"	"+200%"	"+250%"
1	Clothing and Accessory	Service	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
2	High-tech	Textile	High-tech	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
3	Chemical Manufacturing	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation
4	Daily Instrument	Chemical Manufacturing	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
5	Weapon	Raw Manufacturing	Weapon	Weapon	Weapon	Weapon
6	Transportation	High-tech	Transportation	Transportation	Transportation	Transportation
7	Machinery	Clothing and Accessory	Machinery	Machinery	Machinery	Machinery
8	Raw Manufacturing	Machinery	Raw Manufacturing	Textile	Textile	Textile
9	Textile	Transportation	Textile	High-tech	High-tech	High-tech
10	Service	Weapon	Service	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
11	Chemicals	Daily Instrument	Chemical Manufacturing	Service	Service	Service
12	Precious Stone	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
13	Animal and By-product	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
14	Mineral and Stone	Vegetable, Fruit and Plant	Chemicals	Chemicals	Chemicals	Chemicals
15	Wood Product	Chemicals	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
16	Vegetable, Fruit and Plant	Mineral and Stone	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Metal	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product
18	Food and Preparation	Metal	Metal	Metal	Metal	Metal
EfCI	1.911	1.654	1.843	1.839	1.837	1.838

Table 16: Aggregated Food and Preparation 2009 (Incremental Sensitivity)

The Contingency Plan aims to present a set of solutions with respect to upheavals in different industries. Instead of compensating for the declining production directly, the best strategy is to develop those incipiently profitable industries. In this manner, we can derive the contingency strategy for countries to deal with possible abrupt declines of crucial industries. In some cases, the SDV remains unchanged even when a zero-opportunity-gain industry declines to its 20% in terms of export. We can then conclude the industry is not indispensable to the country. As a clarification, the three consecutive N/As means that the country doesn't even need to respond to a collapse in the industry. Countries tend to have distinct solutions to the up and down of the same industry.

4.3 Result and interpretation

The sensitivity analysis of Japan ends up with the results below.

	High-tech	Food and Preparation
Decremental	1.864	0.000
Incremental	0.763	-0.003

Table 17: Sensitivity Score of Japan 2014

	High-tech	Food and Preparation
Decremental	1.483	0.000
Incremental	0.632	0.000

Table 18: Sensitivity Score of Japan 2009

Japan has sensitivity scores typical to developed countries. Japan is barely influenced by the flucuation in Food and Preparation industry, while is subject to the up and down in High-tech. Japan's sensitivity to High Tech industry is obviously lower than the benchmark in 2009, but not 2014.

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Raw Manufacturing	Service
2	Transportation	Transportation	Transportation	Transportation	Daily Instrument	Raw Manufacturing
3	Machinery	Machinery	Machinery	Machinery	Transportation	Daily Instrument
4	Metal	Metal	Metal	Metal	Machinery	Transportation
5	High-tech	High-tech	High-tech	High-tech	Metal	Machinery
6	Chemicals	Chemicals	Chemicals	Chemicals	High-tech	Metal
7	Weapon	Weapon	Weapon	Weapon	Chemicals	Chemicals
8	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
9	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Clothing and Accessory	Weapon	Weapon
10	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Chemical Manufacturing	Clothing and Accessory	Chemical Manufacturing
11	Precious Stone	Precious Stone	Textile	Textile	Precious Stone	Textile
12	Textile	Textile	Precious Stone	Precious Stone	Service	Clothing and Accessory
13	Service	Service	Service	Service	Chemical Manufacturing	Precious Stone
14	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Textile	Animal and By-product
15	Wood Product	Wood Product	Wood Product	Wood Product	Animal and By-product	Wood Product
16	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Wood Product	Vegetable, Fruit and Plant
17	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Vegetable, Fruit and Plant	Food and Preparation
18	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Food and Preparation	High-tech
ECI	2.491	2.491	2.516	2.506	2.282	1.521

Table 19: Japan High Tech 2014 (Decreasing Sensitivity)

Ranking	0	"+50%"	"+100%"	"+150%"	"+200%"	"+250%"
1	Daily Instrument	Daily Instrument	Transportation	Daily Instrument	Daily Instrument	Daily Instrument
2	Transportation	Transportation	Machinery	Machinery	Machinery	Machinery
3	Machinery	Machinery	High-tech	Transportation	Transportation	High-tech
4	Metal	Metal	Chemical Manufacturing	High-tech	High-tech	Transportation
5	High-tech	High-tech	Textile	Chemical Manufacturing	Chemical Manufacturing	Textile
6	Chemicals	Weapon	Weapon	Textile	Textile	Chemical Manufacturing
7	Weapon	Chemicals	Daily Instrument	Raw Manufacturing	Weapon	Raw Manufacturing
8	Mineral and Stone	Chemical Manufacturing	Chemicals	Weapon	Raw Manufacturing	Weapon
9	Chemical Manufacturing	Textile	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
10	Clothing and Accessory	Clothing and Accessory	Raw Manufacturing	Chemicals	Chemicals	Service
11	Precious Stone	Service	Service	Service	Service	Chemicals
12	Textile	Precious Stone	Metal	Metal	Metal	Wood Product
13	Service	Mineral and Stone	Wood Product	Wood Product	Wood Product	Metal
14	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
15	Wood Product	Raw Manufacturing	Precious Stone	Precious Stone	Food and Preparation	Food and Preparation
16	Raw Manufacturing	Wood Product	Food and Preparation	Food and Preparation	Precious Stone	Precious Stone
17	Food and Preparation	Food and Preparation	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
18	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
EfCI	2.491	3.057	3.911	3.799	3.976	4.019

Table 20: Japan High Tech 2014 (Incremental Sensitivity)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
2	Transportation	Transportation	Transportation	Transportation	Transportation	Transportation
3	Machinery	Machinery	Machinery	Machinery	Machinery	Machinery
4	Metal	Metal	Metal	Metal	Metal	Metal
5	High-tech	High-tech	High-tech	High-tech	Textile	Textile
6	Chemicals	Chemicals	Chemicals	Chemicals	High-tech	Chemicals
7	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Chemicals	Raw Manufacturing
8	Weapon	Weapon	Chemical Manufacturing	Chemical Manufacturing	Raw Manufacturing	Chemical Manufacturing
9	Chemical Manufacturing	Chemical Manufacturing	Weapon	Weapon	Chemical Manufacturing	Weapon
10	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Weapon	Clothing and Accessory
11	Service	Textile	Textile	Textile	Clothing and Accessory	Mineral and Stone
12	Textile	Service	Service	Service	Mineral and Stone	Precious Stone
13	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Service	Service
14	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Wood Product
15	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
16	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product	Vegetable, Fruit and Plant
17	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Food and Preparation
18	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	High-tech
EfCI	2.679	2.670	2.670	2.670	2.381	1.919

Table 21: Japan High Tech 2009 (Decreasing Sensitivity)

Ranking	0	"+50%"	"+100%"	"+150%"	"+200%"	"+250%"
1	Daily Instrument	Daily Instrument	Transportation	Transportation	Transportation	Daily Instrument
2	Transportation	Transportation	Machinery	Machinery	Machinery	Transportation
3	Machinery	Machinery	Metal	High-tech	High-tech	High-tech
4	Metal	Metal	High-tech	Daily Instrument	Daily Instrument	Machinery
5	High-tech	High-tech	Weapon	Chemicals	Chemicals	Weapon
6	Chemicals	Chemicals	Chemicals	Weapon	Weapon	Chemicals
7	Raw Manufacturing	Weapon	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Textile
8	Weapon	Chemical Manufacturing	Daily Instrument	Textile	Textile	Chemical Manufacturing
9	Chemical Manufacturing	Textile	Textile	Mineral and Stone	Raw Manufacturing	Raw Manufacturing
10	Mineral and Stone	Mineral and Stone	Mineral and Stone	Raw Manufacturing	Metal	Metal
11	Service	Clothing and Accessory	Raw Manufacturing	Metal	Mineral and Stone	Mineral and Stone
12	Textile	Precious Stone	Clothing and Accessory	Service	Service	Service
13	Clothing and Accessory	Service	Service	Clothing and Accessory	Precious Stone	Wood Product
14	Precious Stone	Raw Manufacturing	Precious Stone	Precious Stone	Clothing and Accessory	Precious Stone
15	Animal and By-product	Wood Product	Wood Product	Wood Product	Wood Product	Clothing and Accessory
16	Wood Product	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Vegetable, Fruit and Plant	Food and Preparation	Food and Preparation	Food and Preparation	Animal and By-product	Food and Preparation
18	Food and Preparation	Animal and By-product	Animal and By-product	Animal and By-product	Food and Preparation	Animal and By-product
EfCI	2.679	3.488	3.849	4.884	4.959	4.977

Table 22: Japan High Tech 2009 (Incremental Sensitivity)

As shown in Table 19---Table 22 above, the economy of Japan is in a much more sensitive and fragile state in 2014, compared to 2009. If the High Tech industry has some surprising breakthroughs, Japan would receive additional stimulations in Machinery and Daily Instrument. In 2014, a decline of High Tech industry would make the SDV of Japan deteriorate to a great extent.

Decreasing export of each industry in Japan's current production space by 60% respectively, we derive the contingency strategy shown below in Table 14.

	1	2	3
Animal and By-product	N/A	N/A	N/A
Vegetable, Fruit and Plant	N/A	N/A	N/A
Food and Preparation	N/A	N/A	N/A
Mineral and Stone	N/A	N/A	N/A
Chemicals	N/A	N/A	N/A
Chemical Manufacturing	N/A	N/A	N/A
Raw Manufacturing	N/A	N/A	N/A
Wood Product	N/A	N/A	N/A
Clothing and Accessory	N/A	N/A	N/A
Textile	N/A	N/A	N/A
Metal	N/A	N/A	N/A
Machinery	N/A	N/A	N/A
Transportation	N/A	N/A	N/A
Weapon	N/A	N/A	N/A
Daily Instrument	N/A	N/A	N/A
Precious Stone	N/A	N/A	N/A
High-tech	Mineral and Stone	N/A	N/A
Service	N/A	N/A	N/A

Table 23: Contingency Table of Japan during Industry Strikes 2014

	1	2	3
Animal and By-product	N/A	N/A	N/A
Vegetable, Fruit and Plant	N/A	N/A	N/A
Food and Preparation	N/A	N/A	N/A
Mineral and Stone	N/A	N/A	N/A
Chemicals	N/A	N/A	N/A
Chemical Manufacturing	N/A	N/A	N/A
Raw Manufacturing	N/A	N/A	N/A
Wood Product	N/A	N/A	N/A
Clothing and Accessory	N/A	N/A	N/A
Textile	N/A	N/A	N/A
Metal	N/A	N/A	N/A
Machinery	N/A	N/A	N/A
Transportation	N/A	N/A	N/A
Weapon	N/A	N/A	N/A
Daily Instrument	N/A	N/A	N/A
Precious Stone	N/A	N/A	N/A
High-tech	N/A	N/A	N/A
Service	N/A	N/A	N/A

Table 24: Contingency Table of Japan during Industry Strikes 2009

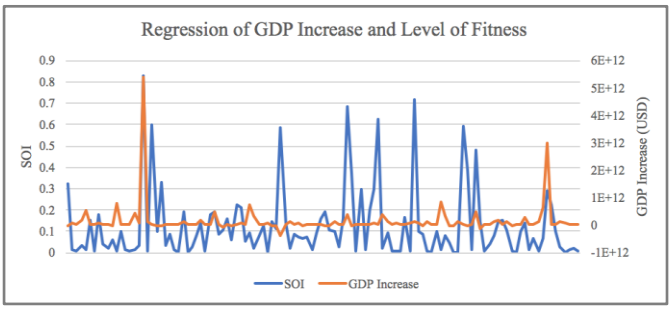
Developing Mineral and Stone is the solution to declines in High tech in 2014. In 2009, however, Japan a extremely stable SDV.

5. Macro-level Predictions

5.1 Structural Optimality Index

Indeed, such choice of prioritization has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our developmental strategies are and how deviations from the suggested developmental path will lead to non-optimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the R-Square obtained from regressing the country's actual development over the suggested development strategy. Both strategies are quantified in nature, with the actual development represented by export increase in major industry groups and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The R-square from the regression thus provides us with a concrete measure of how the development path adopted by the country has fitted with/differed from our suggestion.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which as defined as the net increase in GDP (USD), over their SOI. Figure 6 illustrates the regression output, which shows a strong positive correlation between countries' level of fitness and their five-year GDP growth, significant at the 1% level and achieving an adjusted R^2 of 11.9%. In other words, the more a country develops as suggested by calculations on a complexity level, the better economic outcomes it achieves.



Statistics	Value
Multiple R	0.355
R-Square	0.126
Adjusted R^2	0.119
Variable	**Value**
Coefficient	$1.249*10^{12}$
t-stat	4.061
P-value	0.000(***)

Figure 6: Regression of Five-year GDP Growth over SOI

5.2 SOI of Japan and Its Implications

Japan, for instance, has a relatively low level of structural optimality as compared with its peers. As the regression in Figure 7 suggests (the orange curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), Japan demonstrates a moderate level of correlation with the suggested developmental path, having an adjusted R-square of around 58.3% and an annual growth rate of -1.5%. This is in direct contrast to emerging economies such as China, where there is high SOI of around 82.8% and a corresponding GDP growth of 5.39 billion USD, making it one of the fastest developing economies during the period. In sum, Japan's stagnant economic progress from 2009 to 2014 can be at least partially explained by a mismatched development strategy that demonstrates a considerable level of deviations from our suggested developmental path. In specific, Japan invested heavily in the high-tech industry, while it should have focused more on industry groups such as machinery and service.

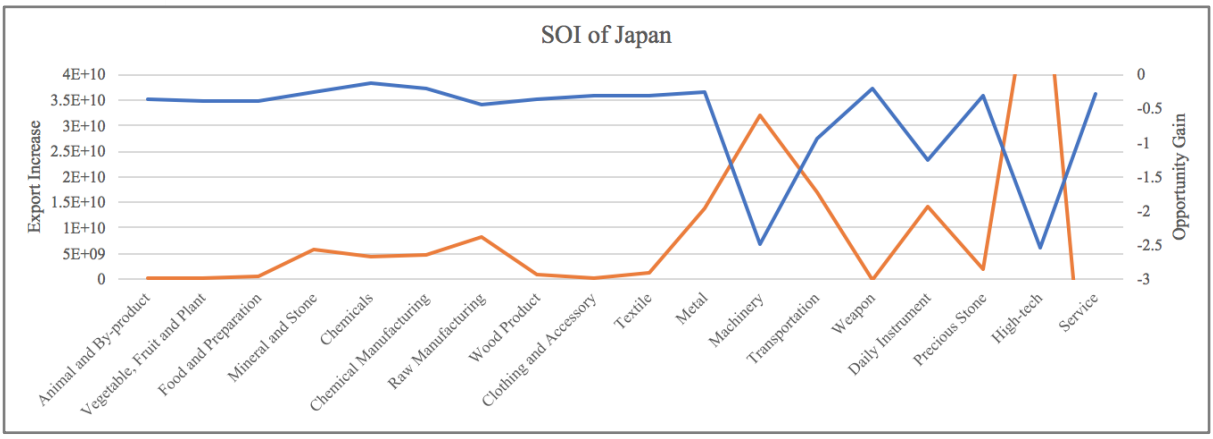


Figure 7: SOI ofJapan (2009-2014)

6. Concluding Remarks

In conclusion, Japan has demonstrated a particularly low level of economic growth in terms of GDP during the period from 2009 to 2014, concurrent with a stagnation in economic complexity as compared with other major Asianor developedeconomies in the world.

The lack of economic progress during the period, we believe, can be partially explained through analysis of the economic and product complexity structure of Japan on the industry group level. The product classification system we adopt allow us to measure the actual development for various industry groups and to observe that Japan aggressively invested in high-tech, machinery and transportation industry groups during the period, while its developed peers and emerging economies around the globe have adopted a slightly more conservative growth path.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, Japan could further its gains by staying optimal with its current industry structure and there is no immediate need of reform because no industry has a positive opportunity gain at its current stage. We further note that Japan has a slightly mismatched development strategy that indicates a moderate level of structural optimality on the complexity level, having an adjusted R-square of around 58.3%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Japan indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure balancing as suggested in the paper.